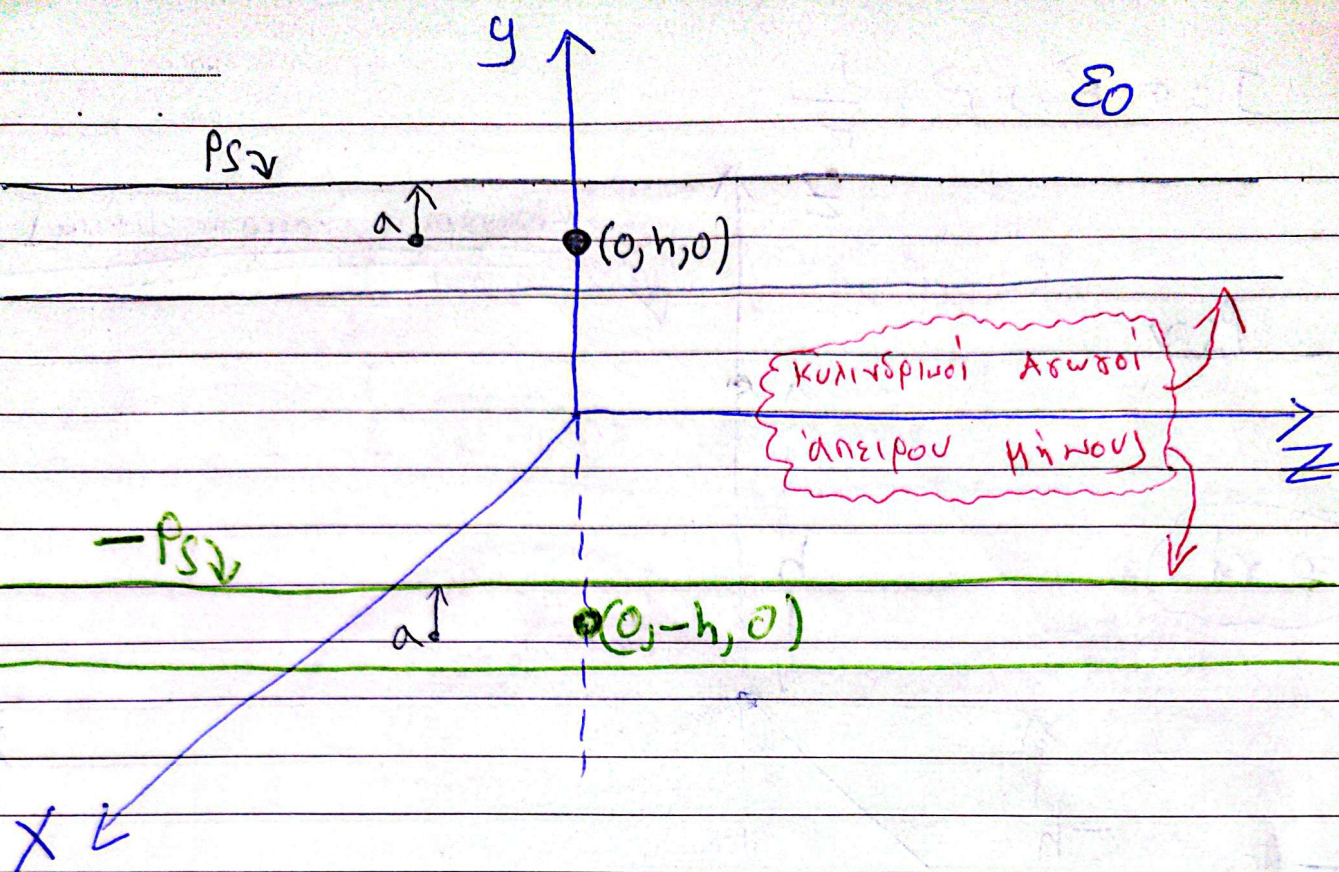




Λύση

$$\vec{r}_2 = (x^0, y^0, 0) - (0, h, 0) \Rightarrow \vec{r}_2 = x^0 \hat{x}_0 + (y^0 - h) \hat{y}_0 + 0 \hat{z}_0$$

$$\vec{r}_2 = x^0 \hat{x}_0 + (y^0 - h) \hat{y}_0, \quad \hat{r}_2 = \frac{x^0 \hat{x}_0 + (y^0 - h) \hat{y}_0}{\sqrt{x^{02} + (y^0 - h)^2}}$$

$$\vec{r}_2 = (x^0, y^0, 0) - (0, -h, 0) \Rightarrow \vec{r}_2 = x^0 \hat{x}_0 + (y^0 + h) \hat{y}_0$$

$$\hat{r}_2 = \frac{x^0 \hat{x}_0 + (y^0 + h) \hat{y}_0}{\sqrt{x^{02} + (y^0 + h)^2}}$$

$$y = h$$

$$h - y = a \Rightarrow y = h - a \quad y - h = a \Rightarrow y = a + h$$

$$|\vec{r}_2| = \sqrt{x^{02} + (y^0 - h)^2}, \quad |\vec{r}_2| = \sqrt{x^{02} + (y^0 + h)^2}$$

$$\cos(\phi_1) = \frac{x^0}{\sqrt{x^{02} + (y^0 - h)^2}}, \quad \sin(\phi_1) = \frac{y^0 - h}{\sqrt{x^{02} + (y^0 - h)^2}}$$

$$\cos(\phi_2) = \frac{x^0}{\sqrt{x^{02} + (y^0 + h)^2}}, \quad \sin(\phi_2) = \frac{y^0 + h}{\sqrt{x^{02} + (y^0 + h)^2}}$$

Ευκλείδης

μΕ

cos θ

sin θ

Gauss για $\rho < a$: $\vec{D} = 0$ (Ακωστός)

Gauss για $\rho > a$: $\int_0^{2\pi} \int_0^L |\vec{D}| \rho \, dz \, d\phi = Q' \cdot \cancel{\rho} = 2\pi \cancel{\rho} |\vec{D}|$

$$\Rightarrow |\vec{D}| = \frac{Q'}{2\pi \rho}$$

Εάν ρ_s σταθερό: $Q' = \rho_s \int_0^{2\pi} \int_0^L \rho \, dz \, d\phi = \rho_s \cdot 2\pi a = Q'$

Άρα $\vec{D} = \frac{2\pi a \rho_s}{2\pi \rho} \hat{\rho}_0 \Rightarrow \vec{D} = \frac{a \rho_s}{\rho} \hat{\rho}_0 \left[\frac{C_6}{m^2} \right] \quad \forall \rho > a$

Κατονομήση

$$\vec{D}_2 = \frac{a \rho_s}{|\vec{\rho}_2|} \hat{\rho}_2 = \frac{a \rho_s}{\sqrt{x^{02} + (y^0 - h)^2}} \cdot \frac{x^0 \hat{x}_0 + (y^0 - h) \hat{y}_0}{\sqrt{x^{02} + (y^0 - h)^2}}$$

$$\vec{D}_2 = \frac{a \rho_s}{x^{02} + (y^0 - h)^2} [x^0 \hat{x}_0 + (y^0 - h) \hat{y}_0] \left[\frac{C_6}{m^2} \right]$$

$$\vec{D}_2 = \frac{-a \rho_s}{x^{02} + (y^0 + h)^2} [x^0 \hat{x}_0 + (y^0 + h) \hat{y}_0] \left[\frac{C_6}{m^2} \right]$$

$$\vec{D} = \vec{D}_1 + \vec{D}_2$$

$$\forall y \geq 0$$

$$\vec{D} = \frac{a \rho_s}{x^2 + (y-h)^2} [x \hat{x}_0 + (y-h) \hat{y}_0] - \frac{a \rho_s}{x^2 + (y+h)^2} [x \hat{x}_0 + (y+h) \hat{y}_0]$$

~~$$\vec{D}$$~~

cherc, da πrerei gia $y=0$ (Eninelo xy)
va βxet hadezo stn επιφάνεια

$$\vec{D}(x, 0) = \frac{a \rho_s}{x^2 + h^2} [x \hat{x}_0 - h \hat{y}_0] - \frac{a \rho_s}{x^2 + h^2} [x \hat{x}_0 + h \hat{y}_0]$$

$$= \cancel{\frac{a \rho_s}{x^2 + h^2} x \hat{x}_0} - \frac{a h \rho_s}{x^2 + h^2} \hat{y}_0 - \cancel{\frac{a \rho_s}{x^2 + h^2} x \hat{x}_0} - \frac{a h \rho_s}{x^2 + h^2} \hat{y}_0$$

$$\vec{D}(x, 0) = \frac{2 a h \rho_s}{x^2 + h^2} (-\hat{y}_0)$$

$$\perp y=0$$

$$\rightarrow > 0$$

κα) προς το $-\hat{y}_0$, λογικο αφού $\rho_s > 0$

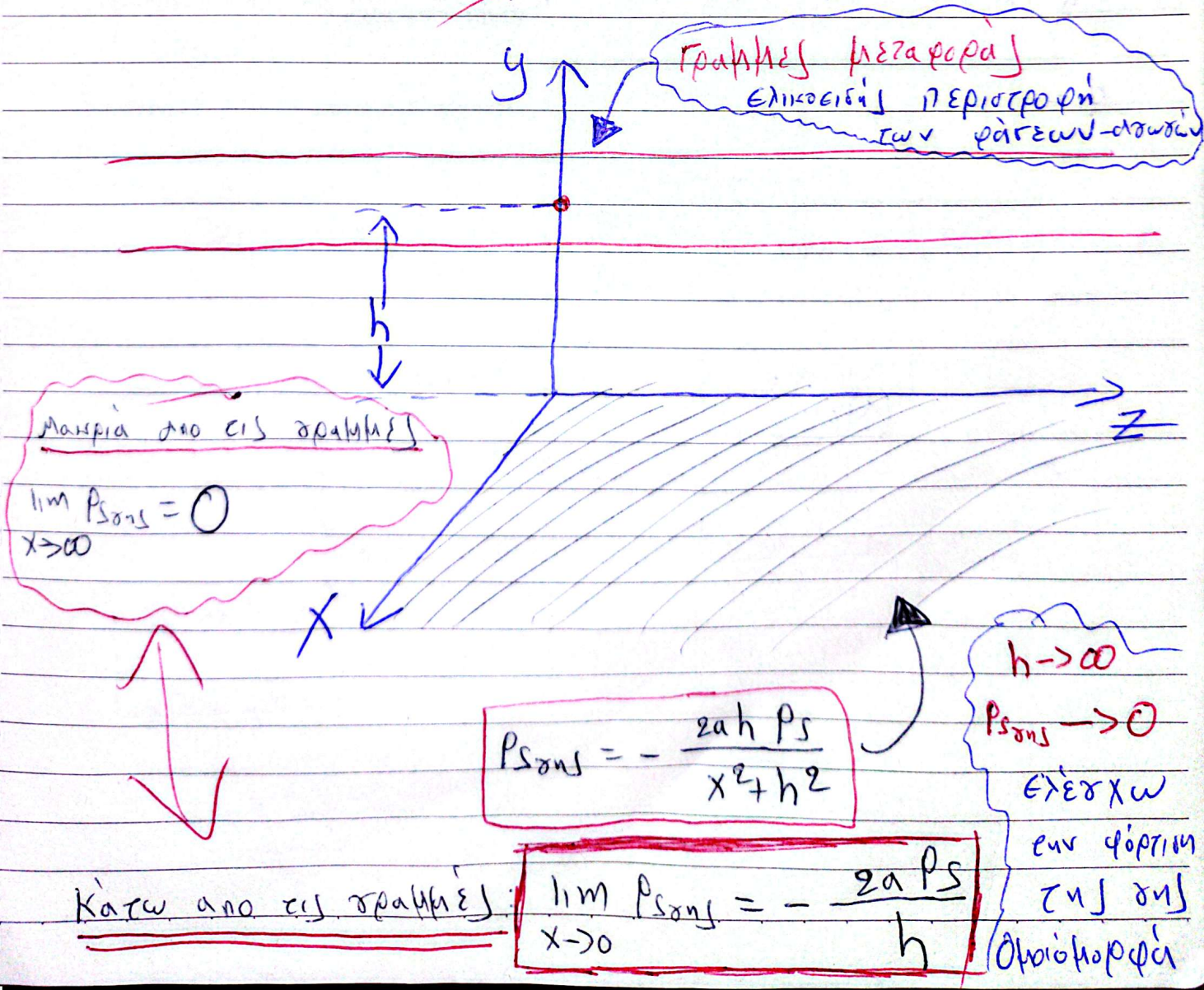
← Για $y < 0 \rightarrow \sigma \sigma \omega \tau \acute{o}$

$$(\vec{D}_2 - \vec{D}_1) \cdot (-\hat{g}_0) \Big|_{y=0} = \rho_{\text{surf}} \Rightarrow \vec{D}_2 \cdot \hat{g}_0$$

$$= \left[\frac{a \rho_s}{x^2 + (y-h)^2} x \hat{x}_0 + \frac{a \rho_s}{x^2 + (y-h)^2} (y-h) \hat{y}_0 - \frac{a \rho_s}{x^2 + (y+h)^2} x \hat{x}_0 - \frac{a \rho_s}{x^2 + (y+h)^2} (y+h) \hat{y}_0 \right] \Big|_{y=0}^{\hat{y}_0}$$

$$\rho_{\text{surf}} = \frac{-ah \rho_s}{x^2 + h^2} - \frac{ah \rho_s}{x^2 + h^2} \Rightarrow \rho_{\text{surf}} = - \frac{2ah \rho_s}{x^2 + h^2} \left[\frac{C_6}{m^2} \right]$$

check $\frac{m \cdot m \cdot \frac{C_6}{m^2}}{m^2} = \frac{C_6}{m^2}$ OK



$$\Gamma_{\text{in}} \quad r \geq a : \varphi_L = \int_{S_L}^{\rho_{\infty}} \frac{a \rho_S}{\epsilon_0 \rho_L} d\rho_L \Rightarrow \varphi_1 = \frac{a \rho_S}{\epsilon_0} \ln \left| \frac{\rho_{\infty}}{\rho_L} \right|$$

$$\varphi_2 = \int_{S_2}^{\rho_{\infty}} -\frac{a \rho_S}{\epsilon_0 \rho_2} d\rho_2 \Rightarrow \varphi_2 = -\frac{a \rho_S}{\epsilon_0} \ln \left| \frac{\rho_{\infty}}{\rho_2} \right|$$

$$\Gamma_{\text{in}} \quad r < a : \varphi_1 = \int_{S_L}^a 0 + \int_a^{\rho_{\infty}} \frac{a \rho_S}{\epsilon_0 \rho_L} d\rho_L \Rightarrow \varphi_1 = \frac{a \rho_S}{\epsilon_0} \ln \left| \frac{\rho_{\infty}}{a} \right|$$

$$\varphi_2 = \int_{S_2}^a 0 + \int_a^{\rho_{\infty}} -\frac{a \rho_S}{\epsilon_0 \rho_2} d\rho_2 \Rightarrow \varphi_2 = -\frac{a \rho_S}{\epsilon_0} \ln \left| \frac{\rho_{\infty}}{a} \right|$$

Εσωτερικά αναλογικά $\boxed{\rho_{\infty} = a}$ ώστε $\ln \frac{a}{a} = 0$

$$\varphi = \varphi_1 + \varphi_2 \Rightarrow \varphi = \frac{a \rho_S}{\epsilon_0} \ln \left| \frac{a}{\rho_L} \right| - \frac{a \rho_S}{\epsilon_0} \ln \left| \frac{a}{\rho_2} \right|$$

$$\varphi = \frac{a \rho_S}{\epsilon_0} \ln \left| \frac{\frac{a}{\rho_L}}{\frac{a}{\rho_2}} \right| = \frac{a \rho_S}{\epsilon_0} \ln \frac{\rho_2}{\rho_L}$$

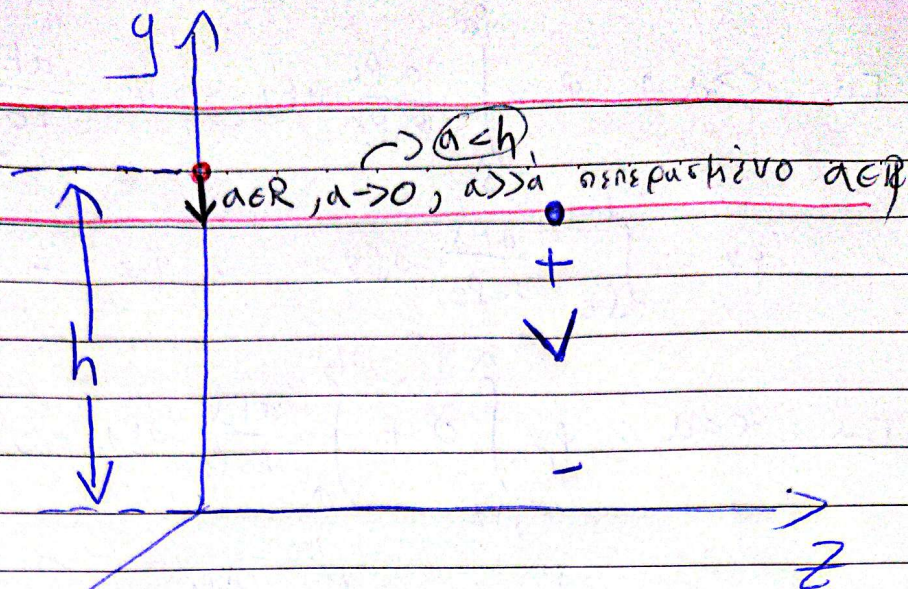
$$\varphi = \frac{a \rho_S}{\epsilon_0} \ln \frac{\sqrt{x^2 + (y+h)^2}}{\sqrt{x^2 + (y-h)^2}} \Rightarrow \varphi = \frac{a \rho_S}{\epsilon_0} \ln \frac{x^2 + (y+h)^2}{x^2 + (y-h)^2} \quad \boxed{V}$$

$$\varphi(0) = \frac{a \rho_S}{\epsilon_0} \ln \frac{x^2 + h^2}{x^2 + h^2} = 0 \quad (\Gamma_{\text{in}})!!!$$

$$\begin{cases} \rho_L = \sqrt{x^2 + (y-h)^2} \Rightarrow a^2 = x^2 + (y-h)^2 \\ \rho_2 = \sqrt{x^2 + (y+h)^2} \Rightarrow a^2 = x^2 + (y+h)^2 \quad (-) \end{cases} \quad \downarrow$$

$$0 = \cancel{y^2 - 2yh + h^2} - \cancel{y^2 - 2yh - h^2} = -4yh = 0 \Rightarrow \boxed{y=0}$$

$$a^2 = x^2 + (0+h)^2 = x^2 + h^2 \Rightarrow \boxed{x^2 = a^2 - h^2}$$



$$V = \varphi(0, h-a) = \frac{a p_s}{\epsilon_0} \ln \sqrt{\frac{0^2 + (h-a+h)^2}{0^2 + (h-a-h)^2}}$$

$$= \frac{a p_s}{\epsilon_0} \ln \frac{\sqrt{(2h-a)^2}}{\sqrt{a^2}} = \frac{a p_s}{\epsilon_0} \ln \left(\frac{2h-a}{a} \right) = V$$

~~For $a \rightarrow 0$, $V = \frac{a p_s}{\epsilon_0} \ln 2$~~

$$p_s = \frac{\epsilon_0 \cdot V}{a \cdot \ln \left(\frac{2h-a}{a} \right)} \left[\frac{C_b}{m^2} \right]$$

check $\frac{\frac{F}{m} \cdot \frac{C_b}{F}}{m} = \frac{C_b}{m^2}$ OK

Είσιw ορατήw η μισw L , τότε

$$Q = p_s \cdot S = p_s \cdot 2\pi a L = \frac{\epsilon_0 V}{a \ln \left(\frac{2h-a}{a} \right)} \cdot 2\pi a L$$

$$\Rightarrow Q = \frac{2\pi L \epsilon_0 \cdot V}{\ln \left(\frac{2h-a}{a} \right)} [C_b]$$

check $\frac{\frac{F}{m} \cdot \frac{C_b}{F}}{L} = C_b$ OK

$$C = \frac{Q}{V} = \frac{\frac{2\pi L \epsilon_0 V}{\ln\left(\frac{2h-a}{a}\right)}}{V} \Rightarrow C = \frac{2\pi L \epsilon_0}{\ln\left(\frac{2h-a}{a}\right)} \left[\frac{F}{m} \right]$$

NO.

Date

$$C' = \frac{C}{L}$$

$$\Rightarrow C' = \frac{2\pi \epsilon_0}{\ln\left(\frac{2h-a}{a}\right)} \left[\frac{F}{m} \right]$$

ηx

$$\epsilon_0 = 8,854 \cdot 10^{-12} \frac{F}{m}, h = 10m, a = 10^{-2}m$$

$$C' = \frac{2\pi \cdot 8,854 \cdot 10^{-12} \frac{F}{m}}{\ln\left(\frac{2 \cdot 10m - 10^{-2}m}{10^{-2}m}\right)} \Rightarrow C' = 7,32 \left[\frac{nF}{m} \right]$$

Χωρίζουμε ως προς την Γm